

Daily Review

Booklet p.15-17 — Forces on Rockets

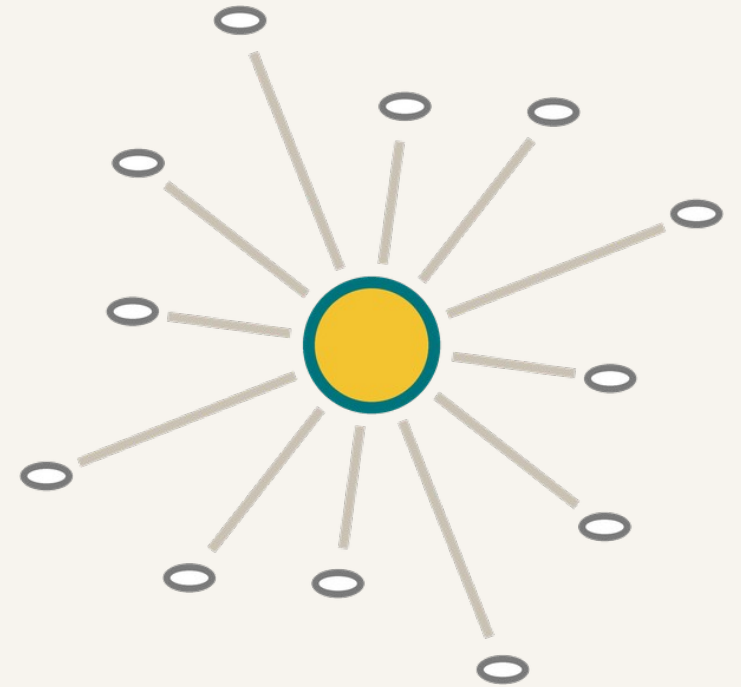
- 1 State Newton's third law.
- 2 A 100 kg probe has an unbalanced force of 30 N. Find a .
- 3 Name the forces on a rocket at take-off.
- 4 What must the thrust be bigger than to lift off?
- 5 What is a Newton pair?

SPACE

The Big Bang

Where did everything come from —
and when?

National 5 Physics (S3)



THE BIG QUESTION

Almost every smudge here is a whole galaxy, and they are all rushing apart. What does that say about the start?



Webb, 2022 — NASA, ESA, CSA, STScI

PART 1

The Big Bang Theory

How the universe began, and how old it is.

By the end of this part...

- 1 I can give a basic description of the Big Bang theory
- 2 I can state the estimated age of the universe

Show me — Describe each word

SHOW ME BOARDS

1 universe

2 galaxy

3 the Big Bang

4 expanding

5 evidence

Show me — Describe each word

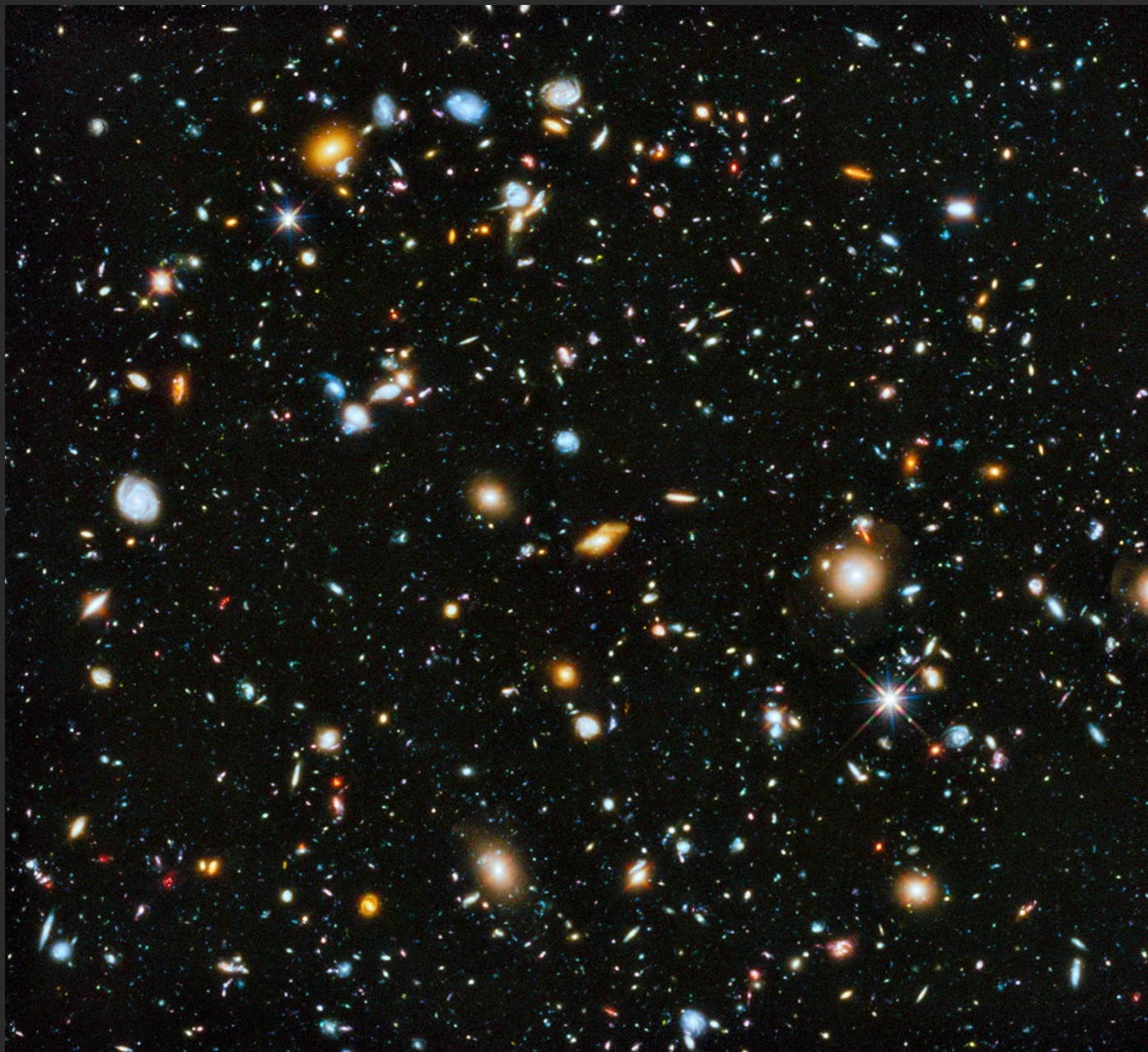
SHOW ME BOARDS

6

redshift

7

microwave background

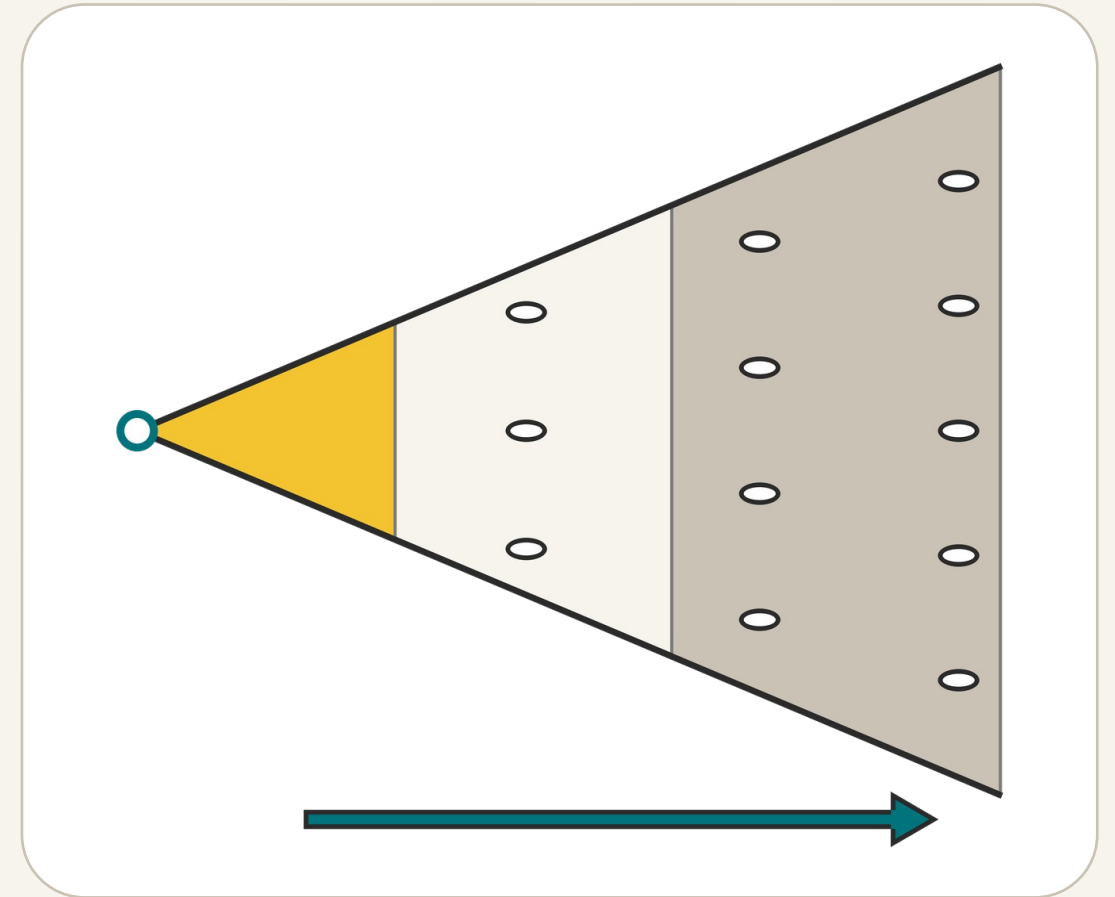


The Hubble Ultra Deep Field — NASA / ESA

The Big Bang

Booklet p.20

- The universe began as a tiny, very hot, very dense point.
- Since then it has been expanding and cooling.
- Galaxies are still moving further apart today.



How old is the universe?

Booklet p.20

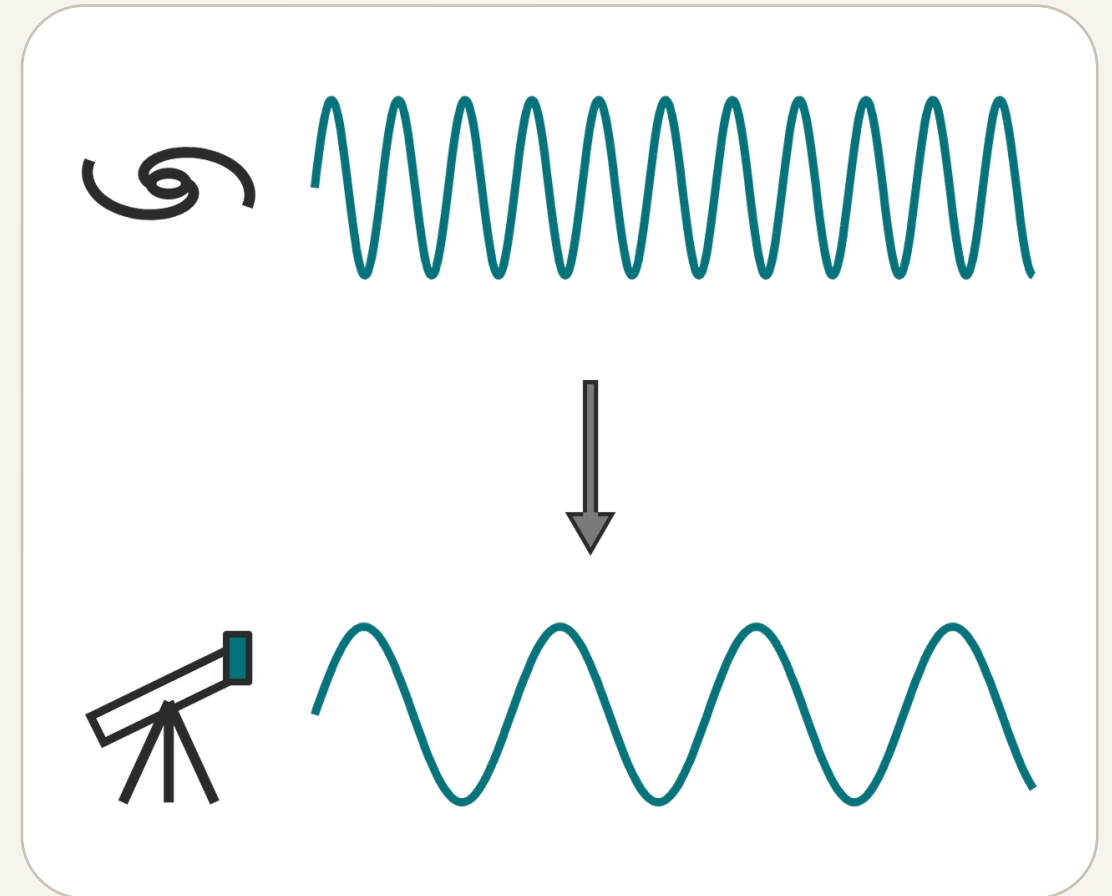
About 13.8 billion years

The estimated time from the Big Bang to today.

Expanding space stretches light

Evidence for the Big Bang

- Space itself expands: the gaps between galaxies grow.
- Light crossing those gaps is stretched with them.
- It arrives with a longer wavelength – redshifted.
- The further away a galaxy is, the redder its light.



Hinge question

FINGER VOTE

Think first — then show A to E on your fingers, 1 to 5

Which of these best describes the Big Bang?

- A** An explosion of a star out in space
- B** Space itself expanding from a tiny hot point
- C** An explosion at the centre of the universe
- D** Matter flying out into empty space
- E** The Earth forming from a cloud of dust.

Show me – the Big Bang

SHOW ME BOARDS

EXTRA PRACTICE

- 1 Is the universe getting hotter or cooler?
- 2 Give two words for the universe at its start.
- 3 Is 13.8 billion years more than 14 million?
- 4 What does the theory describe the origin of?
- 5 Was the universe bigger or smaller in the past?

Agree, disagree or it depends?

SHOW ME BOARDS

Boards up: AGREE, DISAGREE or IT DEPENDS — and why

- 1 The Big Bang was an explosion in space.
- 2 The universe is expanding into something.
- 3 Redshift means the light slows down.
- 4 Light from distant galaxies is redder.
- 5 The universe has a centre.

Task 1 – the age of the universe

Booklet p.20

Complete the definition.

Booklet p.20, Task 1

Task 1 – Answers

ANSWERS

Booklet p.20

- The universe is approximately 13.8 billion years old.

Task 2 – describe the Big Bang

Booklet p.20

Watch the clip about the Big Bang,
then describe the theory.

Booklet p.20, Task 2

Task 2 – Answers

ANSWERS

Booklet p.20

- The universe began as a very small, very hot, very dense point.
- It has been expanding and cooling ever since.

Extension – the evidence

Booklet p.20

Research the three pieces of evidence
and describe each one.

Booklet p.20, Extension

Extension – Answers

ANSWERS

Booklet p.20

Evidence	What it shows
Microwave background	the cooled glow of the early universe, from every direction
Olbers' paradox	the night sky is dark, so the universe has not lasted forever
Light elements	the amounts of hydrogen and helium match the Big Bang

Checkpoint

- 1 Give a basic description of the Big Bang theory.
- 2 State the estimated age of the universe.
- 3 What has the universe been doing since the Big Bang?

Exam practice

Past paper and specimen paper

- National 5 Physics 2024, Section 1, question 8. 1 mark.
- Three statements about the universe — which are correct?
- Specimen paper, Section 1, question 6. 1 mark. The same pattern.
- Then booklet p.21, Task 3.

LOOKING AHEAD

Spectra

Starlight carries a barcode. Reading it tells us what a star is made of.

Plenary

- 1 The big question: what does everything rushing apart tell us?
- 2 Describe the universe at the moment of the Big Bang.
- 3 What has happened to the universe since?
- 4 State the estimated age of the universe.
- 5 Name one piece of evidence for the Big Bang.

- 1 Is the universe expanding or shrinking?
- 2 Why does light from a distant galaxy look redder?
- 3 Is the universe hotter or cooler now than at the start?
- 4 Were galaxies closer together in the past?
- 5 What does the Big Bang theory describe?

Extension

- 1 Why is 'an explosion in space' a misleading picture of the Big Bang?
- 2 Suggest why the Big Bang is called a theory.
- 3 Why can we not see anything older than the universe itself?

Extension

EXTRA PRACTICE

- 1 If galaxies are moving apart now, what must have been true in the past?
- 2 The microwave background comes from every direction. Why is that important?
- 3 Suggest why the night sky would be bright in a never-changing universe.

Course specification

N5 Physics course specification v4.0, p.8 – Cosmology

- Basic description of the ‘Big Bang’ theory of the origin of the universe.
- Knowledge of the approximate estimated age of the universe.